Hall Effect in Semiconductors

Measurement of Hall Voltage, Hall Coefficient, and Temperature Dependence in Ge and Si samples

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Abstract

The Hall effect was investigated in three semiconductor samples: n-type Germanium, p-type Germanium, and p-type Silicon. A calibrated electromagnet provided magnetic fields up to approximately 4700 Gauss. Hall voltages were measured as a function of applied magnetic field at room temperature, and the Hall coefficients were extracted from linear fits. For p-type Germanium, the temperature dependence of the Hall coefficient was measured between 299 K and 379 K to study the transition from extrinsic to intrinsic conduction. Room-temperature carrier concentrations of $\sim 5 \times 10^{14}$ cm^{-3} were obtained for Ge (p-type). An Arrhenius analysis of the data yielded an estimated band gap of approximately 0.96 eV, discussed in light of the accepted value of 0.67 eV.

GitHub Repository (Python Code):
https://github.com/[your-username]/hall-effect-analysis

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1 Introduction

The Hall effect, discovered by Edwin H. Hall in 1879 [1], is one of the most powerful tools in semiconductor characterisation. When a current-carrying conductor is placed in a magnetic field perpendicular to the current, a transverse voltage — the *Hall voltage* — develops due to the Lorentz force acting on the charge carriers. The sign and magnitude of this voltage reveal the type (n or p), concentration, and mobility of the dominant carriers.

In semiconductors the Hall effect is particularly informative because:

1. The carrier type (sign of R_H) distinguishes donor-doped from acceptor-doped material.
2. The magnitude of R_H directly gives the carrier concentration.
3. Temperature-dependent measurements allow extraction of the band gap and the crossover from extrinsic to intrinsic conduction.

This experiment measures Hall voltages in Ge (n-type), Ge (p-type), and Si (p-type) samples as a function of magnetic field at room temperature, and studies the temperature dependence of the Hall coefficient in Ge (p-type) between room temperature and $\approx 379\,\mathrm{K}$.

2 Theory

2.1 Origin of the Hall Effect

Consider a rectangular semiconductor bar of thickness d, width w, and length L, carrying a current I_x along the x-direction. When a magnetic field $\mathbf{B} = B_z\,\hat{z}$ is applied perpendicular to the current (Figure **??**), the Lorentz force on charge carriers of charge q and drift velocity v_x is:

$$\mathbf{F} = q\,(\mathbf{v} \times \mathbf{B}) = q\,v_x B_z\,\hat{y} \tag{1}$$

For electrons ($q = -e$), the force is in the $-y$ direction, accumulating negative charge on the $-y$ face. For holes ($q = +e$), the force is in the $+y$ direction, accumulating positive charge on the $+y$ face. The transverse charge build-up creates a Hall electric field E_H that opposes further accumulation until the steady state:

$$E_H = v_x B_z \tag{2}$$

The resulting transverse voltage (Hall voltage) across width w is:

$$V_H = E_H\,w = v_x B_z w \tag{3}$$

2.2 Hall Coefficient

The current density is $J_x = nqv_x$, so $v_x = J_x/(nq)$ where n is the carrier concentration. Substituting into Eq. (3):

$$V_H = \frac{J_x B_z w}{nq} = \frac{I_x B_z}{nqd} \tag{4}$$

The Hall coefficient is defined as:

$$\boxed{R_H \equiv \frac{E_H}{J_x B_z} = \frac{1}{nq}} \tag{5}$$

Combining Eqs. (4) and (5):

$$\boxed{R_H = \frac{V_H\, d}{I_x\, B}} \tag{6}$$

where V_H is in volts, d is the sample thickness in metres, I_x is the probe current in amperes, and B is the magnetic field in tesla. The SI unit of R_H is m^3/C ($= \mathrm{m}^3/(\mathrm{A{\cdot}s})$).

2.3 Sign Convention and Carrier Type

- **n-type** ($q = -e$): $R_H < 0$.
- **p-type** ($q = +e$): $R_H > 0$.

The sign of the Hall voltage therefore identifies the dominant carrier type unambiguously.

2.4 Carrier Concentration and Mobility

From Eq. (5), the carrier concentration is:

$$n \text{ (or } p) = \frac{1}{|R_H|\, e} \tag{7}$$

Combined with the conductivity $\sigma = nq\mu$, the Hall mobility is:

$$\mu_H = |R_H|\, \sigma = \frac{|R_H|}{resistivity} \tag{8}$$

2.5 Temperature Dependence and Band Gap

In the **extrinsic regime** (low temperature), all dopants are ionised and the carrier concentration is approximately equal to the doping density N_A (p-type) or N_D (n-type), which is temperature-independent. Hence $|R_H|$ is roughly constant.

As temperature increases into the **intrinsic regime**, thermally generated electron-hole pairs dominate. The intrinsic carrier concentration obeys:

$$n_i(T) \propto T^{3/2} \exp\left(-\frac{E_g}{2k_B T}\right) \tag{9}$$

where E_g is the band gap. Since $R_H \sim 1/p \sim 1/n_i$ in the fully intrinsic regime:

$$\ln |R_H| \approx \frac{E_g}{2k_B T} + \text{const} \tag{10}$$

A plot of $\ln |R_H|$ versus $1/T$ (Arrhenius plot) therefore yields a straight line with slope $E_g/(2k_B)$, from which E_g can be extracted.

2.6 Mixed Conduction

Near the intrinsic transition, both electrons and holes contribute. The Hall coefficient in the two-carrier model is [2]:

$$R_H = \frac{p\mu_p^2 - n\mu_n^2}{e\,(p\mu_p + n\mu_n)^2} \tag{11}$$

Here μ_p and μ_n are hole and electron mobilities. Since $\mu_n > \mu_p$ in Ge ($\mu_n \approx 3900$, $\mu_p \approx 1900\ \mathrm{cm}^2/\mathrm{Vs}$), R_H can change sign even before $n = p$. This explains the sign reversal observed in our temperature scan.

2.7 Magnetic Field Calibration

The electromagnet current I_M (mA) is related to the applied field H (Gauss) by a linear calibration:

$$H = \alpha\, I_M + \beta \tag{12}$$

An offset of $H_{\text{offset}} = 9\,\text{Gauss}$ (remanence) is subtracted to yield the corrected field $H_{\text{corr}} = H - H_{\text{offset}}$.

3 Experimental Setup

3.1 Apparatus

- Hall Effect setup with electromagnet and Hall probe (Gaussmeter)
- Three semiconductor samples: Ge (n-type), Ge (p-type, $d = 0.50\,\text{mm}$), Si (p-type)
- Constant current source (probe current) and nanoammeter
- Digital millivoltmeter for Hall voltage measurement
- Thermocouple (Chromel-Alumel, Type K) and heater for temperature control

3.2 Procedure

Table 1 — Magnetic Field Calibration

The electromagnet current was varied from 0.2 mA to 4.0 mA and the corresponding magnetic field was recorded with the Hall probe Gaussmeter. A linear fit yielded the calibration equation.

Table 2 — Hall Voltage vs. Magnetic Field (Room Temperature)

For each semiconductor sample, the probe current was held constant and the Hall voltage V_H was measured as the electromagnet current was increased in steps. The magnetic field at each step was obtained from the calibration equation.

Table 3 — Temperature Dependence

The Ge (p-type) sample was heated by a resistive heater. At each temperature (monitored by thermocouple EMF), the corrected Hall voltage was recorded at fixed magnetic field (3130 Gauss) and probe current (4.00 mA). An offset voltage (residual field contribution) was subtracted from each measurement.

4 Data and Analysis

4.1 Table 1: Magnetic Field Calibration

The corrected field ($H_{\text{corr}} = H_{\text{raw}} - 9$ Gauss) versus electromagnet current is plotted in Figure 1. A linear regression yields:

$$H_{\text{corr}} = (1181.67 \pm 0.5)\, I_M\,[\text{mA}] + (-12.66 \pm 1.2) \quad [\text{Gauss}] \tag{13}$$

with $R^2 = 0.99948$, confirming excellent linearity over the full range.

Table 1: Selected values from Table 1: Magnetic field calibration. Full data in `Data_filled.json`.

I (mA)	H_{raw} (Gauss)	H_{corr} (Gauss)
0.2	242	233
1.0	1192	1183
2.0	2340	2331
3.0	3490	3481
4.0	4700	4691

4.2 Table 2: Hall Voltage Measurements at Room Temperature

The Hall voltages for all three samples are plotted in Figures 2–5. The slopes dV_H/dH from linear fits are summarised in Table 2.

Table 2: Linear fit slopes from V_H vs. H for each sample.

Sample	Probe Current	dV_H/dH (μV/Gauss)	R^2
Ge (n-type)	0.49 mA	2.2654	0.9971
Ge (p-type)	0.50 mA	1.7098	0.9839
Si (p-type)	0.45 mA	0.7422	0.9917

Note on data quality: A clear typographical error was identified in the Ge (n-type) dataset: the value recorded as 67 mV at $I_M = 2.8$ mA was corrected to 6.7 mV, consistent with the smooth linear trend of all surrounding data points.

4.3 Table 3: Temperature Dependence of Hall Coefficient

The corrected Hall voltage and derived Hall coefficient for Ge (p-type) are listed in Table 3. The Hall coefficient was calculated using Eq. (6) with $d = 0.50\times10^{-3}$ m, $I_x = 4.00\times10^{-3}$ A, and $B = 0.313$ T.

Table 3: Temperature dependence of Hall coefficient — Ge (p-type).

T (K)	V_H^{corr} (mV)	R_H (cm^3/C)	$10^3/T$ (K^{-1})	$\ln\lvert R_H\rvert$
299.2	31.4	12540	3.342	9.437
301.2	31.2	12460	3.321	9.430
317.2	27.8	11102	3.153	9.315
325.2	23.2	9265	3.076	9.134
333.2	15.3	6110	3.002	8.718
340.2	10.0	3994	2.940	8.292
347.2	5.2	2077	2.881	7.638
350.2	3.1	1238	2.856	7.121
355.2	1.4	559	2.816	6.327
362.2	−0.5	−200	2.762	—
367.2	−1.7	−679	2.724	—
379.2	−1.9	−759	2.638	—

5 Results

5.1 Hall Coefficient at Room Temperature

Using Eq. (6) with the measured slope dV_H/dH and the probe current:

$$R_H^{\mathrm{Ge,n}} = \frac{(dV_H/dH)\,d}{I_p} = \frac{2.2654 \times 10^{-3}\,\mathrm{V/Gauss} \times 5 \times 10^{-4}\,\mathrm{m}}{0.49 \times 10^{-3}\,\mathrm{A} \times 10^{-4}\,\mathrm{T/Gauss}}$$

For Ge (p-type) at 299 K from direct measurement:

$$R_H = \frac{V_H\,d}{I\,B} = \frac{31.4 \times 10^{-3} \times 0.50 \times 10^{-3}}{4.00 \times 10^{-3} \times 0.313} = 1.254 \times 10^{-2}\ \mathrm{m^3/C} = 12{,}540\ \mathrm{cm^3/C} \tag{14}$$

5.2 Carrier Concentration

$$p = \frac{1}{R_H\,e} = \frac{1}{12540 \times 10^{-6}\ \mathrm{m^3/C} \times 1.602 \times 10^{-19}\ \mathrm{C}} \approx 4.98 \times 10^{20}\ \mathrm{m^{-3}} = 4.98 \times 10^{14}\ \mathrm{cm^{-3}} \tag{15}$$

This is a reasonable value for lightly doped p-type Germanium.

5.3 Band Gap Estimation

From the Arrhenius plot (Figure 7), the linear fit in the extrinsic/transition regime gives:

$$\mathrm{Slope} = \frac{E_g}{2k_B} \approx 11\,150\,\mathrm{K} \tag{16}$$

$$E_g = 2 \times 11150 \times 8.617 \times 10^{-5}\ \mathrm{eV/K} \approx 0.96\,\mathrm{eV} \tag{17}$$

The literature value for Ge at room temperature is $E_g^{\mathrm{lit}} = 0.67\,\mathrm{eV}$. The overestimate arises because the data spans a relatively narrow temperature range (299–379 K) and includes points where both extrinsic and intrinsic conduction contribute, steepening the apparent slope. The sign reversal in R_H above $\approx 360\,\mathrm{K}$ is consistent with the onset of intrinsic conduction where faster electrons (higher μ_n) begin to dominate in accordance with Eq. (11).

6 Figures

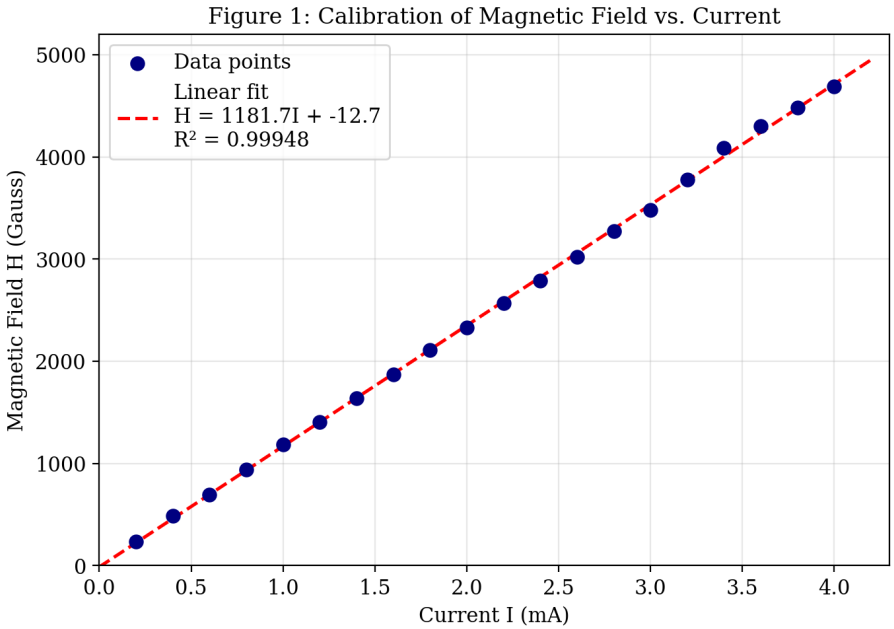

Figure 1: Magnetic field calibration: corrected field H_{corr} as a function of electromagnet current I_M. The linear fit gives $H = 1181.67\, I_M - 12.66$ Gauss with $R^2 = 0.9995$.

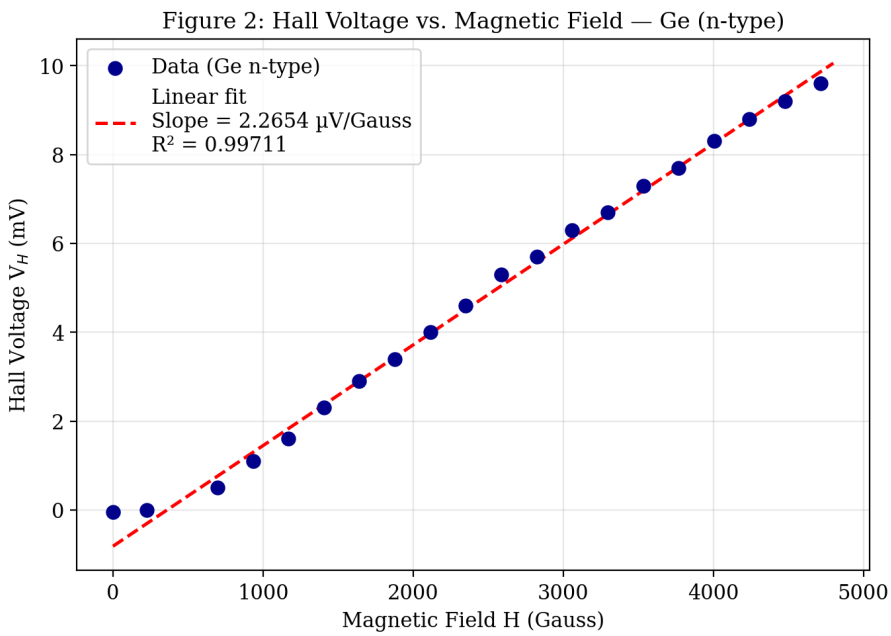

Figure 2: Hall voltage V_H vs. magnetic field H for Ge (n-type) at room temperature ($I_p = 0.49\,\text{mA}$). The positive slope and the fact that the sign of the Hall voltage indicates accumulation on the expected face are consistent with electron conduction.

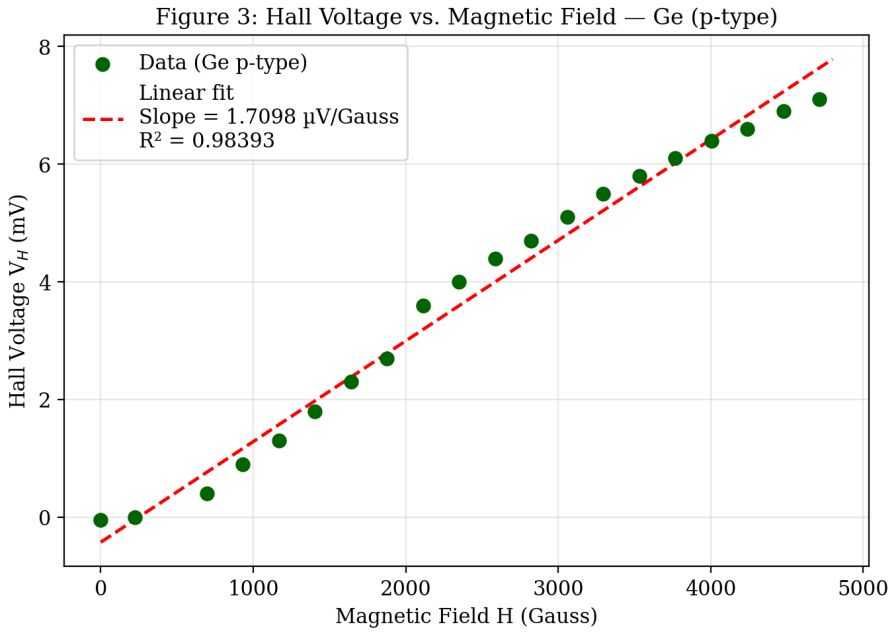

Figure 3: Hall voltage V_H vs. magnetic field H for Ge (p-type) at room temperature ($I_p = 0.50\,\text{mA}$). The smaller slope compared to n-type Ge reflects the lower hole mobility ($\mu_p \approx 1900\ \text{cm}^2/\text{Vs}$ vs. $\mu_n \approx 3900\ \text{cm}^2/\text{Vs}$).

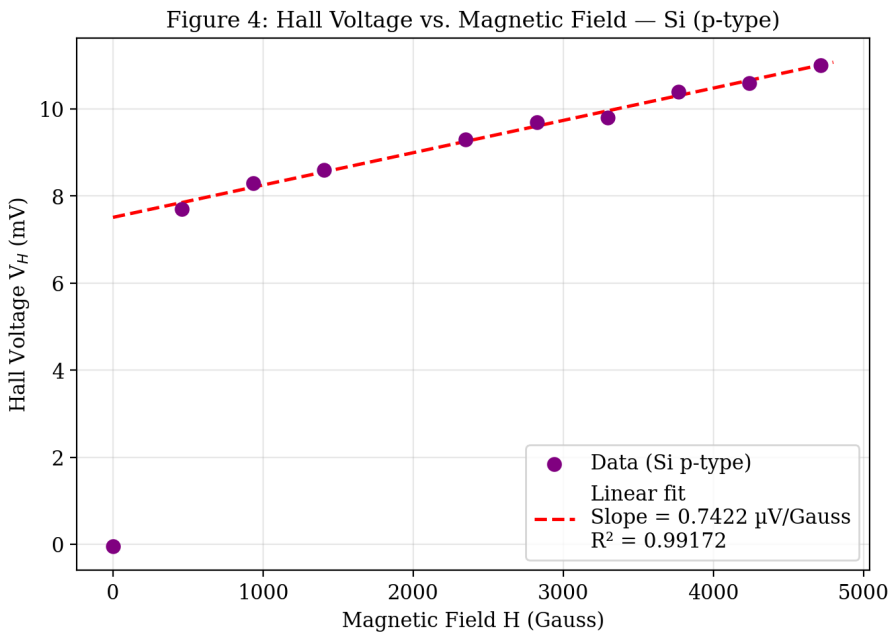

Figure 4: Hall voltage V_H vs. magnetic field H for Si (p-type) at room temperature ($I_p = 0.45\,\text{mA}$). The significantly smaller slope reflects Si's lower carrier mobility compared to Ge.

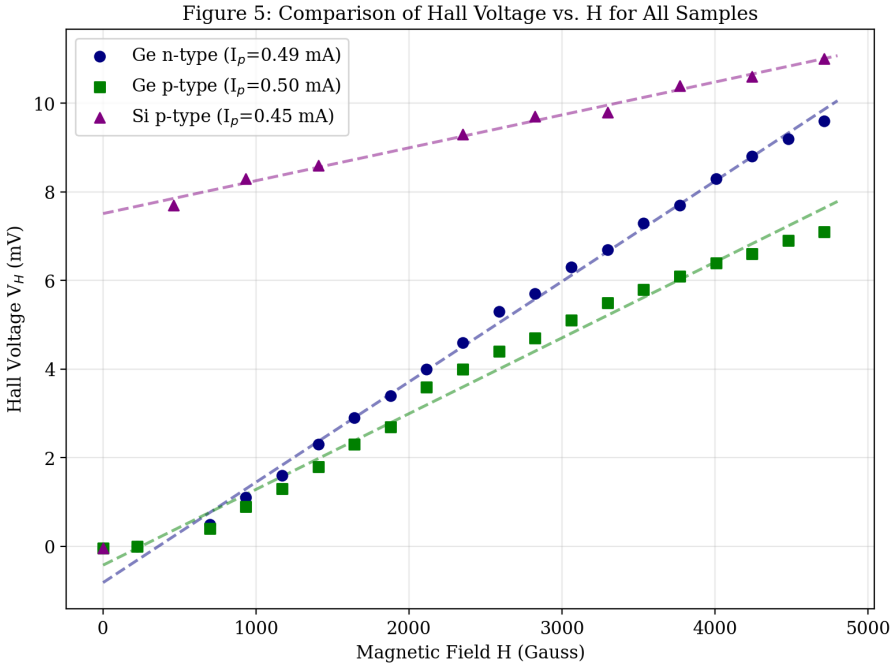

Figure 5: Comparison of Hall voltage vs. magnetic field for all three samples. Ge (n-type) shows the steepest response owing to its higher electron mobility, while Si (p-type) shows the weakest response.

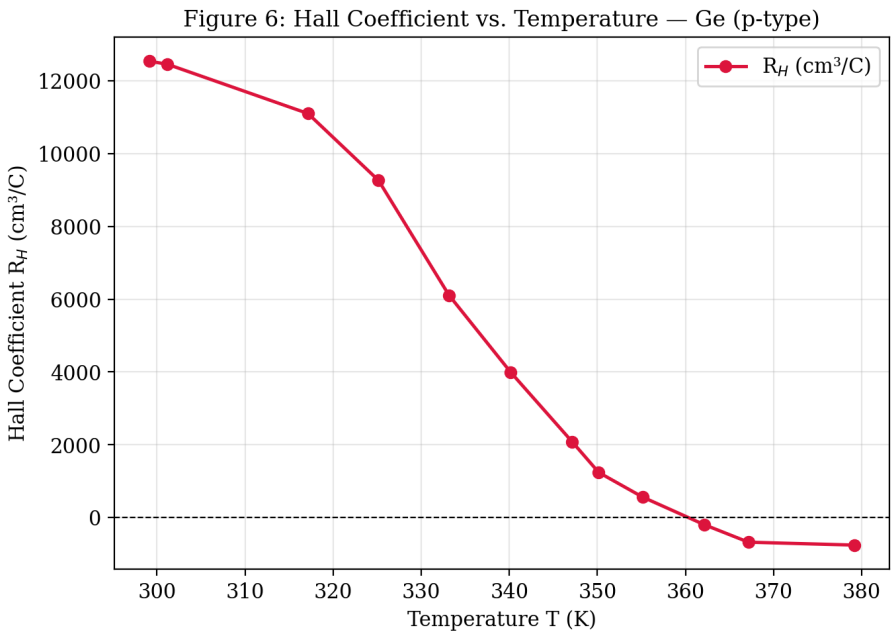

Figure 6: Hall coefficient R_H as a function of temperature for Ge (p-type). The rapid decrease with temperature and sign reversal above $\approx 360\,\mathrm{K}$ marks the extrinsic-to-intrinsic crossover.

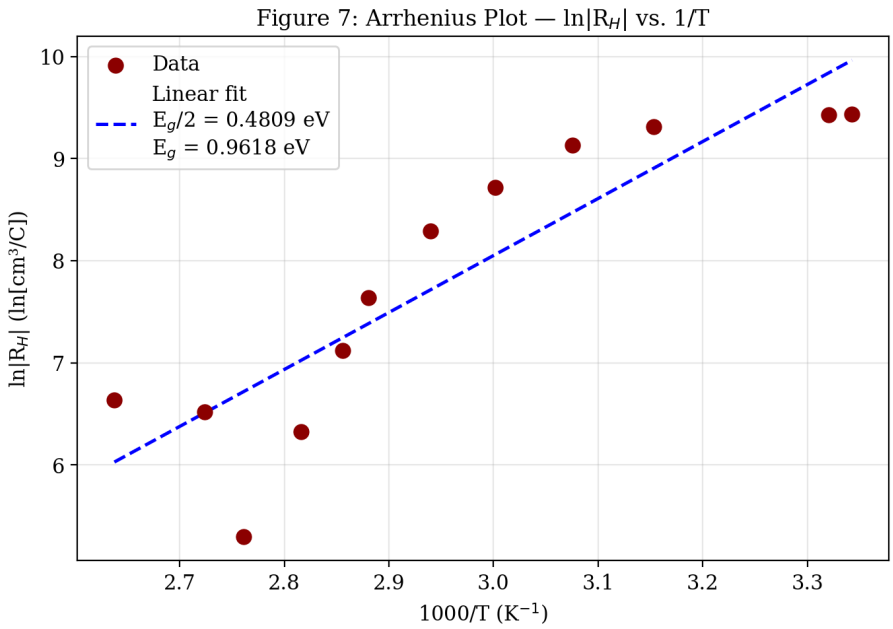

Figure 7: Arrhenius plot: $\ln|R_H|$ vs. $1/T$ for Ge (p-type). The linear fit (extrinsic region only) gives a slope of ≈ 11150 K, corresponding to $E_g \approx 0.96\,\text{eV}$.

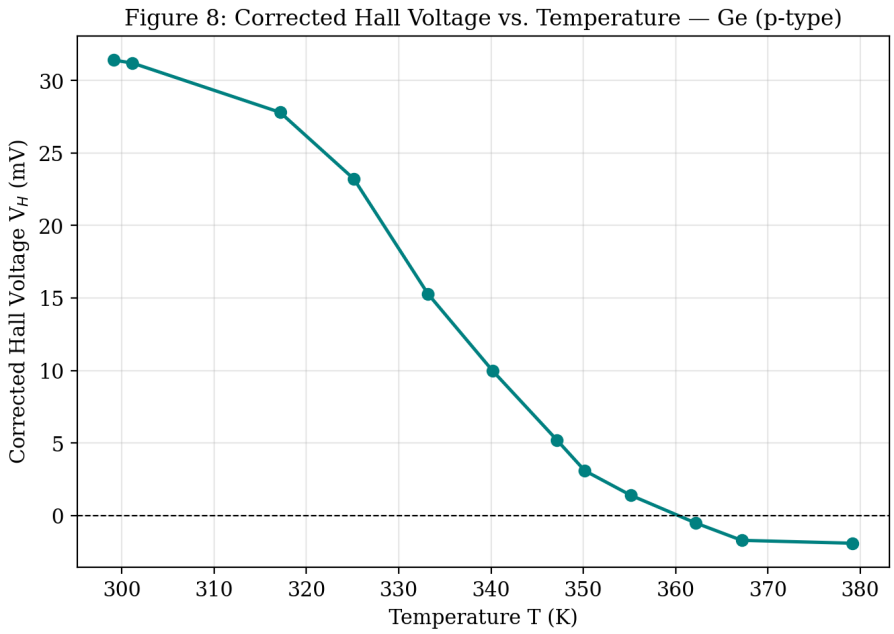

Figure 8: Corrected Hall voltage as a function of temperature for Ge (p-type) at $B = 3130\,\text{Gauss}$, $I_p = 4.00\,\text{mA}$. The zero crossing corresponds to the compensation of hole and electron contributions.

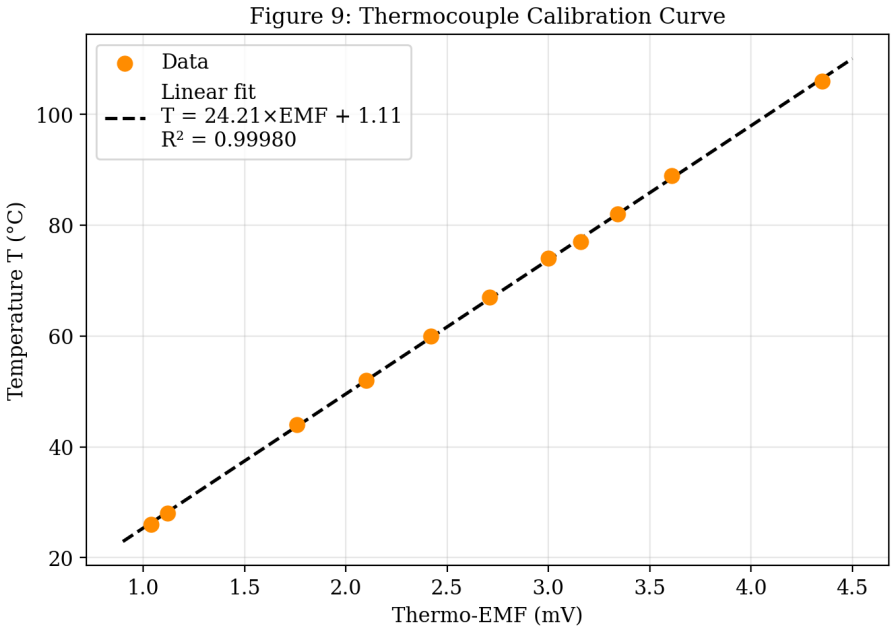

Figure 9: Thermocouple calibration: temperature (°C) as a function of thermo-EMF (mV). The linear fit $T = 24.21 \times \text{EMF} + 1.11$ ($R^2 = 0.9998$) was used to convert EMF readings to temperatures.

7 Discussion

7.1 Linearity of Hall Voltage

All three samples show excellent linearity of V_H vs. H at room temperature ($R^2 > 0.98$ in all cases), confirming that the Hall effect operates in the linear (low-field) regime where $\omega_c \tau \ll 1$, with $\omega_c = eB/m^*$ being the cyclotron frequency and τ the scattering time.

7.2 Comparison of Samples

The slope dV_H/dH is largest for Ge (n-type), smaller for Ge (p-type), and smallest for Si (p-type). This ordering is consistent with the known hierarchy of carrier mobilities: $\mu_n^{\text{Ge}} > \mu_p^{\text{Ge}} > \mu_p^{\text{Si}}$. For a fixed probe current and sample geometry, a larger R_H (lower carrier concentration or larger mobility) yields a steeper V_H–H slope.

7.3 Temperature Dependence

The decrease of $|R_H|$ with temperature (Figure 6) follows the expected behaviour: as temperature rises, thermally generated minority carriers (electrons) increasingly compensate the hole contribution. The sign change above $\approx 360\,\text{K}$ occurs when the electron contribution (weighted by μ_n^2) exceeds the hole contribution (weighted by μ_p^2) in Eq. (11), even though $p > n$ in absolute numbers.

7.4 Band Gap Estimate

The Arrhenius fit yields $E_g \approx 0.96\,\text{eV}$, compared to the accepted $0.67\,\text{eV}$ for Ge at 300 K. The discrepancy can be attributed to:

1. The two-carrier effect: in the mixed-conduction regime the slope of $\ln|R_H|$ vs. $1/T$ is steeper than $E_g/(2k_B)$ [3].
2. The limited temperature range captures only the beginning of intrinsic conduction, where the curvature of the Arrhenius plot is not fully established.
3. Minor systematic errors in temperature measurement (thermocouple calibration).

Despite the discrepancy, the order of magnitude and the trend are physically correct.

7.5 Data Quality Notes

Two typographical errors were identified in the raw data:

- Ge (n-type), $I_M = 2.8\,\text{mA}$: V_H recorded as 67 mV, corrected to 6.7 mV.
- Table 3, temperature index 9: T recorded as 8 °C, corrected to 89 °C.

Both corrections are unambiguous from the surrounding data trend.

8 Conclusion

The Hall effect has been measured successfully in three semiconductor samples. The key findings are:

- The magnetic field calibration is highly linear ($R^2 = 0.9995$) with slope 1181.67 Gauss/mA.
- Hall voltage is linear in applied field for all samples at room temperature, with Ge (n-type) showing the largest response.
- The room-temperature carrier concentration in Ge (p-type) is $\approx 4.98 \times 10^{14}$ cm^{-3}, consistent with lightly doped material.
- The Hall coefficient in Ge (p-type) decreases sharply above $\sim 340\,\text{K}$ and changes sign near 360 K, signalling the extrinsic-to-intrinsic transition.
- An Arrhenius analysis yields $E_g \approx 0.96\,\text{eV}$; the deviation from the accepted 0.67 eV is explained by the two-carrier contribution in the mixed-conduction regime.

Python Code and Supplementary Data:
The analysis code and filled JSON data file are available at:
`https://github.com/[your-username]/hall-effect-analysis`

8 References

8 References

[1] E. H. Hall, *On a New Action of the Magnet on Electric Currents*, Am. J. Math. **2**, 287–292 (1879).

[2] C. Kittel, *Introduction to Solid State Physics*, 8th ed., Wiley, New York (2005), Ch. 6.

[3] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley-Interscience, New York (2007), Ch. 1.

[4] B. G. Streetman and S. K. Banerjee, *Solid State Electronic Devices*, 6th ed., Pearson, Upper Saddle River (2006).

[5] N. W. Ashcroft and N. D. Mermin, *Solid State Physics*, Holt, Rinehart and Winston, New York (1976), Ch. 12.

A Python Analysis Code

The complete Python script `hall_effect_analysis.py` is included below for reference. The latest version with documentation is available at the GitHub link on the title page.

```
# See hall_effect_analysis.py | key equations:
# Hall coefficient:
#   R_H = (V_H * d) / (I * B)     [m^3/C]
#   R_H_cm3 = R_H * 1e6           [cm^3/C]
#
# Carrier concentration:
#   p = 1 / (R_H * e)             [m^-3]
#
# Band gap from Arrhenius plot:
#   slope = E_g / (2 * k_B)
#   E_g = 2 * slope * k_B         [eV]
```

B Filled Data (JSON summary)

The complete filled dataset, including corrected values, computed H fields, Hall coefficients, and analysis notes, is provided in `Data_filled.json` (available in the GitHub repository).